MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL, MODULAR, AND
COMPONENT PARTS
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-C-28754C, dated 13 December 1982, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.3.4.3, delete and substitute:

"3.3.4.3 Contact plating.

"3.3.4.3.1 Overall plating. All contacts shall be gold plated in accordance with Type II, Grade C, Class 1 of MIL-G-45204 to a thickness of 50 microinches minimum. Contacts shall have a minimum nickel underplating of 50 microinches in accordance with Class 1 of QQ-N-290.

"3.3.4.3.2 Localized finish. When specified in the individual specification sheet, a localized finish may be used as an alternate to the overall plating specified in 3.3.4.3.1 (see 3.1)."

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*4.1.1, line 5: Delete "MIL-C-45662" and substitute "MIL-STD-45662".

PAGE 13

*4.5, line 2: Delete "(see 6.5)" and substitute "(see 6.3)".

4.5.1.1, line 2: Delete "... shall be subjected 40 ..." and substitute "... shall be subjected to ...".

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*TABLE IV, subgroup IV: Delete "Insulation resistance (pre-humidity)" in its entirety.

*4.6.2.1, line 1: Between "inspection" and "shall", add "shall be conducted every 12 months on connectors for which retention of qualification is desired and ..."

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*FIGURE 6, NOTE 5: Delete "MIL-STD-10" and substitute "ANSI B46.1-78".

PAGE 33

*Following page 33, add new appendix as printed on pages 2 through 7 of this amendment.

FSC 5935

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10 SCOPE

10.1 Scope. This appendix details the procedure for submission of samples, with related data, for qualification inspection of connectors, contacts, and associated back panel covered by this specification. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30 SUBMISSION

30.1 Samples. All samples shall be submitted to DESC. If extended qualification is desired, the number of tested samples indicated in MIL-C-28754 or this appendix shall be submitted. Also, an equal number of samples that have not been subjected to testing shall be submitted and shall be accompanied by three copies of the test report. Submission of untested samples qualified by extended qualification is not desired, but may be required by the qualifying activity or their agent.

30.1.1 Type III sample submission. Test plate samples shall be submitted in accordance with 4.5.1.2 of MIL-C-28754. Test plates shall be subjected to the test of table IV of MIL-C-28754. Ten of each other wrappost length (see table I of applicable specification sheet) to be qualified shall be submitted as loose parts. The loose parts shall be subjected to visual and mechanical tests (see 4.7.1). Loose part qualification is contingent upon satisfactory results from test plate testing.

30.1.1.1 MIL-C-28754/34 samples. Test plate samples shall be pinned with M28754/34-04 wrappost pins (an alternate pin from table I is permitted if /34-04 wrappost pins are not available).

30.1.1.2 MIL-C-28754/35, /36, and /38 samples. Test plate samples shall be pinned in one 40-pin minimum pattern with one style contact from table I with 9 rows of /34 as applicable.

30.1.1.3 MIL-C-28754/37 samples. Test plate samples shall be pinned at the ends of all 40-pin minimum patterns with one style polarizing bushing set from table I of /37. Ten of each style polarizing bushing (table I, a through s) and ten each of other wrappost lengths (table I) to be qualified shall be submitted as loose parts.

30.1.1.4 MIL-C-38754/39 keying peg. Ten of each keying peg (table I) to be qualified shall be submitted as loose parts. The loose parts shall be subjected to visual and mechanical tests (see 4.7.1) (exclude test plate testing).

30.1.2 Type III group submissions.

30.1.2.1 MIL-C-28754/34. When qualified with /37, the ends of all 40-pin minimum patterns shall contain /37 pins and polarizing bushings. When qualified with /35, /36, and /38, one 40-pin minimum pattern shall be pinned for each /35, /36, or /38 (in place of /34) requiring qualification.

30.1.2.2 MIL-C-28754/35. When qualified with /37, the ends of all 40-pin minimum patterns shall contain /37 pins and polarizing bushings. When qualified for /35 only, all 40-pin minimum patterns shall contain /35 pins, when qualified with /34, (and /36 or /38), one 40-pin minimum pattern of /35 shall be pinned for each /36 or /38 (in place of /34).

30.1.2.3 MIL-C-28754/36. When qualified with /37, the ends of all 40-pin minimum patterns shall contain /37 pins and polarizing bushings. When qualified for /36 only, all 40-pin minimum patterns shall contain /36 pins. When qualified for /34 (and /35 or /38), one 40-pin minimum pattern of /36 shall be pinned and one 40-pin minimum pattern shall be pinned for each /35 or /38 (in place of /34).

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APPENDIX

30.1.2.4 MIL-C-28754/38. When qualified with /37, the ends of all 40-pin minimum patterns shall contain /37 pins and polarizing bushings. When qualified for /38 only, all 40-pin minimum patterns shall contain /38 pins. When qualified with /34 (and /35 or /36), one 40-pin minimum pattern of /38 shall be pinned in place of /34 and one 40-pin minimum pattern shall be pinned for each /35 or /36 (in place of /34).

30.1.3 Type IV and V sample submission reduced testing and extended qualification (see table VIII).

TABLE VIII. Sample submission data.

Military part number M28754/	Perform all tests except	Perform all tests	Delete all tests except	Sample quantity	Extends qualification to M28754/
6-01	Subgroup IV Connector fin torque, fin catilever load, guide rib strength.	N/A	N/A	12	N/A
6-02		N/A	N/A	12	N/A
6-03		N/A	N/A	12	6-01, -02, -04
6-04		N/A	N/A	12	6-01, -02, -03
7-02	N/A	X	N/A	12	6-02
7-03	N/A	X	N/A	12	10-01
7-04	N/A	X	N/A	12	7-02; 6-01, -02, -03, -04
7-05	N/A	X	N/A	12	7-03; 10-01, -03
8-01	Subgroup II Pin shield retention. Subgroup IV Pin shield retention, connector fin torque, fin catilever load, guide rib strength.	N/A	N/A	12	N/A
9-01	N/A	X	N/A	12	11-01
9-02	N/A	X	N/A	12	9-01, -03; 11-01, -02
9-03	N/A	X	N/A	12	11-01, -02
10-01	Subgroup IV Connector fin torque, fin catilever load, guide rib strength.	N/A	N/A	12	N/A
10-02		N/A	N/A	12	N/A
10-03		N/A	N/A	12	10-04
10-04		N/A	N/A	12	N/A
11-01	Subgroup IV Connector fin torque, fin catilever load, guide rib strength.	N/A	N/A	12	N/A
11-02		N/A	N/A	12	11-01
12-01	N/A	X	N/A	12	11-01; 12-02
12-02	N/A	X	N/A	12	11-01; 12-01

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TABLE VIII. Sample submission data - Continued.

Military part number M28754/	Perform all tests except	Perform all tests	Delete all tests except	Sample quantity	Extends qualification to M28754/
12-03	N/A	X	N/A	12	12-01, -02, -04, -05; 11-01, 11-02
12-04	N/A	X	N/A	12	11-01, -02; 12-01, -02, -03, -05
12-05	N/A	X	N/A	12	11-01, -02
13-01	Subgroup IV	N/A	N/A	12	6-02
13-02	Guide rib strength.	N/A	N/A	12	6-04, -02; 13-01
14-01	Same as /8	N/A	N/A	12	N/A
15-01	N/A	N/A	Subgroup I crimp or solder joint resistance. Subroup V testing	20	N/A
15-02	N/A	N/A		20	N/A
16-01	N/A	N/A	Subroup I Visual and mechanical examination.	12	N/A
16-02	N/A	N/A		12	N/A
17-01	N/A	X	N/A	12	15-01; 16-01; 49-01
17-02	N/A	X	N/A	12	15-01; 16-01; 17-01; 49-01, 49-02
17-03	N/A	X	N/A	12	15-01; 16-01; 17-01, -02; 49-01, -02, 03
17-04	N/A	X	N/A	12	15-02; 16-02; 49-01
18-01	N/A	N/A	Subroup I Visual and mechanical examination.	12	N/A
18-02	N/A	N/A		12	N/A
18-03	N/A	N/A		12	N/A
19-01	N/A	N/A	Subroup I Visual and mechanical examination.	12	N/A
19-02	N/A	N/A		12	N/A
19-03	N/A	N/A		12	N/A
20-01	N/A	N/A	Subroup I Visual and mechanical examination.	12	N/A
20-02	N/A	N/A		12	N/A
20-03	N/A	N/A		12	N/A
21-01	N/A	N/A	Subgroup I Visual and mechanical examination.	12	21-02, -03
21-02	N/A	N/A		12	N/A
21-03	N/A	N/A		12	N/A
22-01	N/A	N/A	Subgroup II Vibration, pin shield retention, visual and mechanical examination. Subgroup III Salt spray, visual and mechanical examination. Subgroup IV Shock, pin shield re- tention,	12	22-02, -03
22-02	N/A	N/A		12	N/A
22-03	N/A	N/A		12	N/A
23-01	N/A	N/A		12	23-02, -03
23-02	N/A	N/A		12	N/A
23-03	N/A	N/A		12	N/A

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TABLE VIII. Sample submission data - Continued.

Military part number M28754/	Perform all tests except	Perform all tests	Delete all tests except	Sample quantity	Extends qualification to M28754/
21, 22, 23 continued			Connector fin torque, fin cantilever load, guide rib strength, and visual and mechanical examination.		
24-01	N/A	N/A	Subroup I	12	24-04
24-02	N/A	N/A	Visual and	12	24-05
24-03	N/A	N/A	mechanical	12	24-06
24-04	N/A	N/A	examination.	12	24-01
24-05	N/A	N/A		12	24-02
24-06	N/A	N/A		12	24-03
31-01	Subgroup II	N/A	N/A	6	
31-02	Vibration and	N/A	N/A	N/A	
31-03	these tests that follow vibration. Contact retention, contact resistance, pin shield retention.	N/A	N/A	N/A	1/ 31-02, -03, -04, -05, -07, -08, -09; 50-01, -02, -03, -04;
31-04	Subgroup IV shock, pin	N/A	N/A	N/A	51-01, -02, -03, -04
31-05	shield retention, connector fin	N/A	N/A	N/A	
31-06	torque, guide	N/A	N/A	6	
31-07	rib strength,	N/A	N/A	N/A	
31-08	contact re-	N/A	N/A	N/A	
31-09	tention.	N/A	N/A	N/A	
32-01	N/A	X	N/A	N/A	
32-02	N/A	X	N/A	N/A	31-05, -06, -08,
32-11	N/A	X	N/A	N/A	-09; 32-01, -02,
32-12	N/A	X	N/A	N/A	-11, -12, -14,
32-14	N/A	X	N/A	N/A	-15, -20, -22,
32-15	N/A	X	N/A	N/A	-23; 50-02;
32-20	N/A	X	N/A	N/A	52-01
32-21	N/A	X	N/A	12	
32-22	N/A	X	N/A	N/A	
32-23	N/A	X	N/A	N/A	
33-01	N/A	X	N/A	N/A	2/
33-07	N/A	X	N/A	6	31-01, -02, -03,
33-08	N/A	X	N/A	N/A	-04, -05, -06,
33-09	N/A	X	N/A	N/A	-07, -08, -09;
33-10	N/A	X	N/A	N/A	33-01, -08, -09, -10, -11, -13, -14, -15; 50-02, -03, -04

See footnotes at end of table.

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TABLE VIII. Sample submission data - Continued.

Military part number M28754/	Perform all tests except	Perform all tests	Delete all tests except	Sample quantity	Extends qualification to M28754/
33-11	N/A	X	N/A	N/A	2/ 50-02, -03, -04;
33-12	N/A	X	N/A	6	51-02, -03, -04;
33-13	N/A	X	N/A	N/A	53-01
33-14	N/A	X	N/A	N/A	
33-15	N/A	X	N/A	N/A	
40-01	N/A	N/A	Subgroup I	12	N/A
40-02	N/A	N/A	Visual and	12	N/A
40-03	N/A	N/A	mechanical examination.	12	N/A
41-01	N/A	N/A	Subgroup I	12	N/A
41-02	N/A	N/A	Visual and	12	N/A
41-03	N/A	N/A	mechanical	12	N/A
41-04	N/A	N/A	examination.	12	N/A
42-01	Same as /6	N/A	N/A	12	N/A
43-01	N/A	X	N/A	12	42-01
44-01	Same as /6	N/A	N/A	12	44-02
44-02		N/A	N/A	12	44-01
45-01	N/A	N/A	Same as /21	12	N/A
46-01	N/A	X	N/A	12	44-02; 45-01
47-01	Same as /6	N/A	N/A	12	N/A
48-01	N/A	X	N/A	12	10-02
48-02	N/A	X	N/A	12	48-01; 10-02, -04
49-01	N/A	N/A	Same as /21	12	N/A
49-02	N/A	N/A		12	49-01
49-03	N/A	N/A		12	49-01, -02
50-01	Same as /31	N/A	N/A	12	N/A
50-02		N/A	N/A	12	50-03, -04
50-03		N/A	N/A	12	50-02, -04
50-04		N/A	N/A	12	50-02, -03
51-01	Same as /31	N/A	N/A	12	N/A
51-02		N/A	N/A	12	51-03, -04
51-03		N/A	N/A	12	51-02, -04
51-04		N/A	N/A	12	51-02, -03

See footnotes at end of table.

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TABLE VIII. Sample submission data - Continued.

Military part number M28754/	Perform all tests except	Perform all tests	Delete all tests except	Sample quantity	Extends qualification to M28754/
52-01	N/A	N/A	Same as /21	12	N/A
53-01	N/A	N/A	Same as /21	12	N/A
54-01 54-02	N/A N/A	N/A N/A	Subroup I Visual and mechanical examination and tests called out on specification sheet.	4 4	N/A N/A

1/ To exercise the extended qualification option 6 samples of M28754/31-01 and 6 samples of M28754/31-06 shall be tested for a total of 12 samples.

2/ To exercise the extended qualification option 6 samples of M28754/33-07 and 6 samples of M28754/33-12 shall be tested.

N/A = Not applicable.
X = Affirmative.

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